
CSI
COMPUTER SOCIETY OF INDIA
SFIT
COMPUTER SOCIETY OF INDIA
IN COLLABORATION WITH
GOOGLE DEVELOPER GROUPS
HACKX 2.0
Code for
भारत 5.0

HackX 2.0

Problem Statement Title

EV BHARAT : AI POWERED EV-INTELLIGENCE

Track Name

Future of Mobility & Transportation

Team Name

CodeQuest

Problem Statement

The rapid adoption of Electric Vehicles (EVs) in India has outpaced the availability of intelligent, integrated systems required for safe, efficient, and adaptive mobility. Current EV solutions are fragmented, with battery management, navigation, safety monitoring, and user assistance operating as disconnected components with limited interoperability.

Existing systems fail to incorporate critical real-time factors such as battery health (SoH), energy consumption, traffic complexity, weather variability, and road safety risks—challenges that are particularly pronounced in India's diverse and high-density environments. Moreover, essential capabilities including driver fatigue detection, accident prevention and response, multilingual accessibility, and intuitive user assistance remain inadequate. There is also a lack of simulation-based learning to prepare users for intelligent and semi-autonomous mobility.This gap necessitates a unified, AI-driven mobility intelligence platform that integrates battery analytics, energy-aware routing, real-time safety monitoring, multilingual voice assistance, and simulation capabilities. Such a system aims to enhance efficiency, improve safety outcomes, and enable inclusive, scalable mobility aligned with the vision of Bharat 5.0.

Proposed Solution

An AI-driven unified platform integrating battery analytics, energy-efficient routing, real-time safety monitoring, and multilingual voice assistance to enable safer, smarter, and efficient EV mobility.

Innovation / USP	What Makes This Different?
AI-Based Battery SoH Prediction	Real-Time AI Intelligence
Intelligent EV Routing	Battery+Traffic+Weather Routing
AI Safety (Drowsiness + Accident)	Proactive Safety System
Voice RAG Assistant	Multilingual Accessibility
Simulation Learning	Learning + Prediction Loop
Auto FIR Enforcement	Law Enforcement Integration
V2X + Hardware Integration	Connected Mobility Network

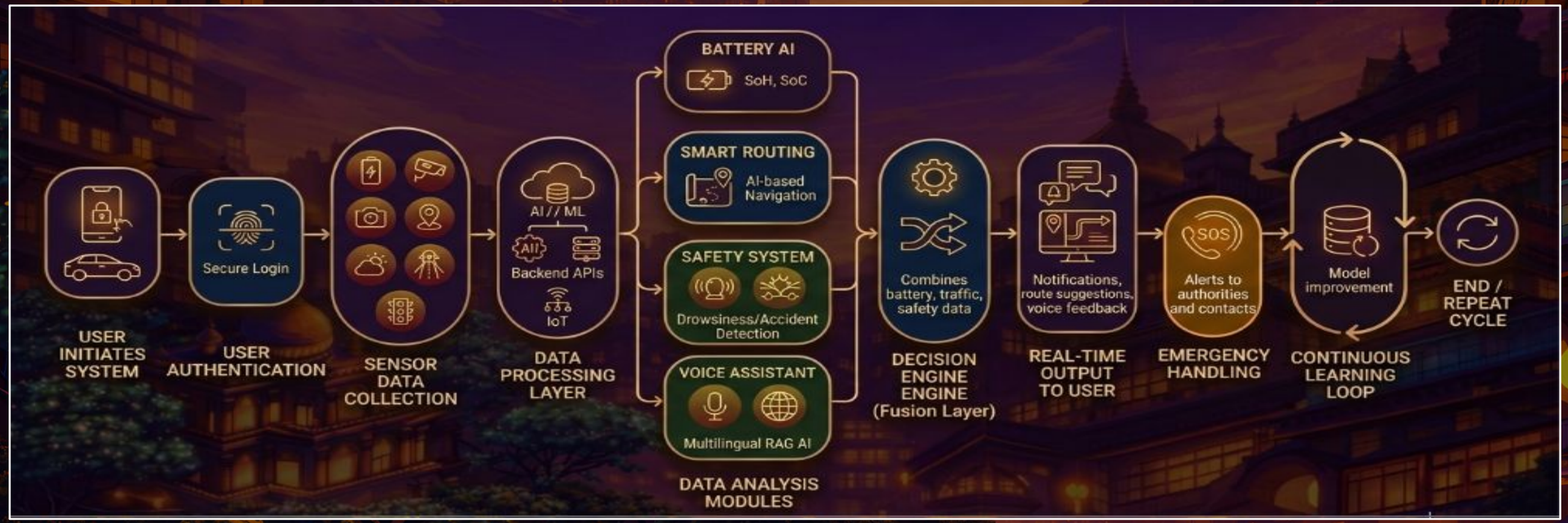
Technical Approach

Technologies Used:

- Frontend-React.js(TypeScript),Vite,Tailwind CSS,PWA & Service Workers,Three.js.
- Backend-Python,FastAPI ,REST_APIs, Pydantic,JW, Authentication,LangChain,Langgraph,Redis,Node.js, SUMO,Docker,C++.
- Database- SQL ,IndexedDB.
- Machine Learning Module- Python, Scikit-learn, Pandas, Numpy , Tensorflow, OpenCV, YOLO.

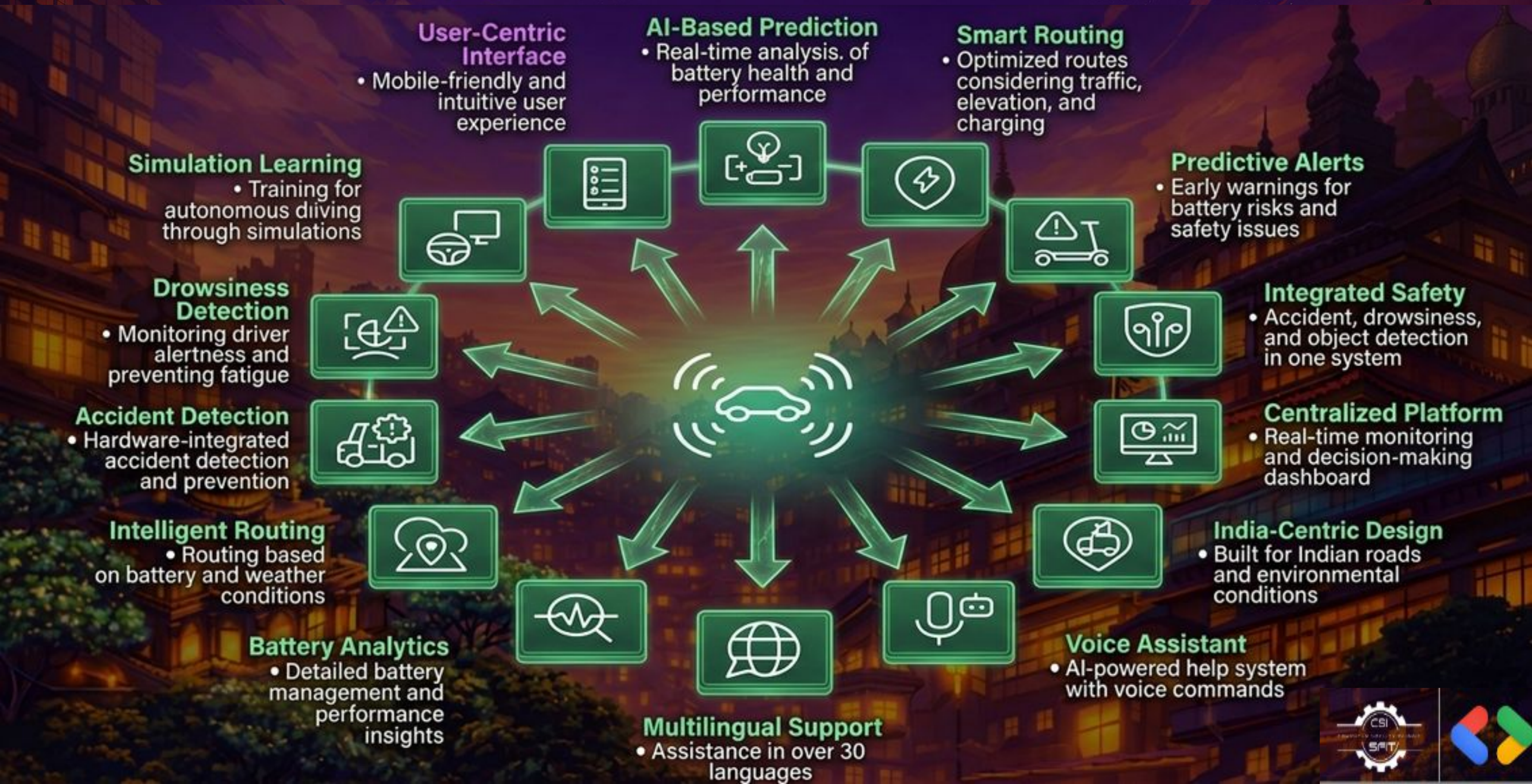
Prototype Link: https://ev-bharat-2txw-eight.vercel.app/

Process Flow



1/

Key Features



Impact & Future Scope

IMPACT

- **Enhanced Safety:** Accident detection, drowsiness monitoring & instant alerts
- **Battery Reliability:** Real-time analysis with predictive insights
- **Inclusive Access:** Multilingual (25+) voice-based AI assistant
- **Smart Documentation:** Centralized license, insurance & policy management
- **Faster Enforcement:** Automatic FIR generation for rapid response

KEY BENEFITS

- **Unified AI Platform:** Battery analytics + routing + safety + assistance
- **Energy-Aware Routing:** Optimized using battery, traffic & weather data
- **Proactive Safety System:** Prevention, detection & real-time alerts
- **Voice-Based RAG Assistant:** Hands-free, intelligent interaction
- **Simulation Learning:** Safe training for real-world driving scenarios
- **Hardware + V2X Ready:** Real-time connectivity with demo capability

SCALABILITY

- **Cloud-Native & Modular:** Easy deployment and expansion
- **Multi-Vehicle Support:** 2W, 3W, 4W & fleet ecosystems
- **Pan-India Adaptability:** Works across urban & rural Indian environments
- **AI-Driven Evolution:** Improves continuously with real-world data

FUTURE ENHANCEMENTS

- **Smart City Integration:** Traffic, infrastructure & emergency systems
- **Advanced V2X Ecosystem:** Vehicle ↔ infrastructure ↔ network communication
- **Autonomous Intelligence:** AI-driven driving decision systems
- **Edge AI Processing:** Low-latency, offline-capable intelligence
- **Sustainability Layer:** Carbon tracking & energy optimization